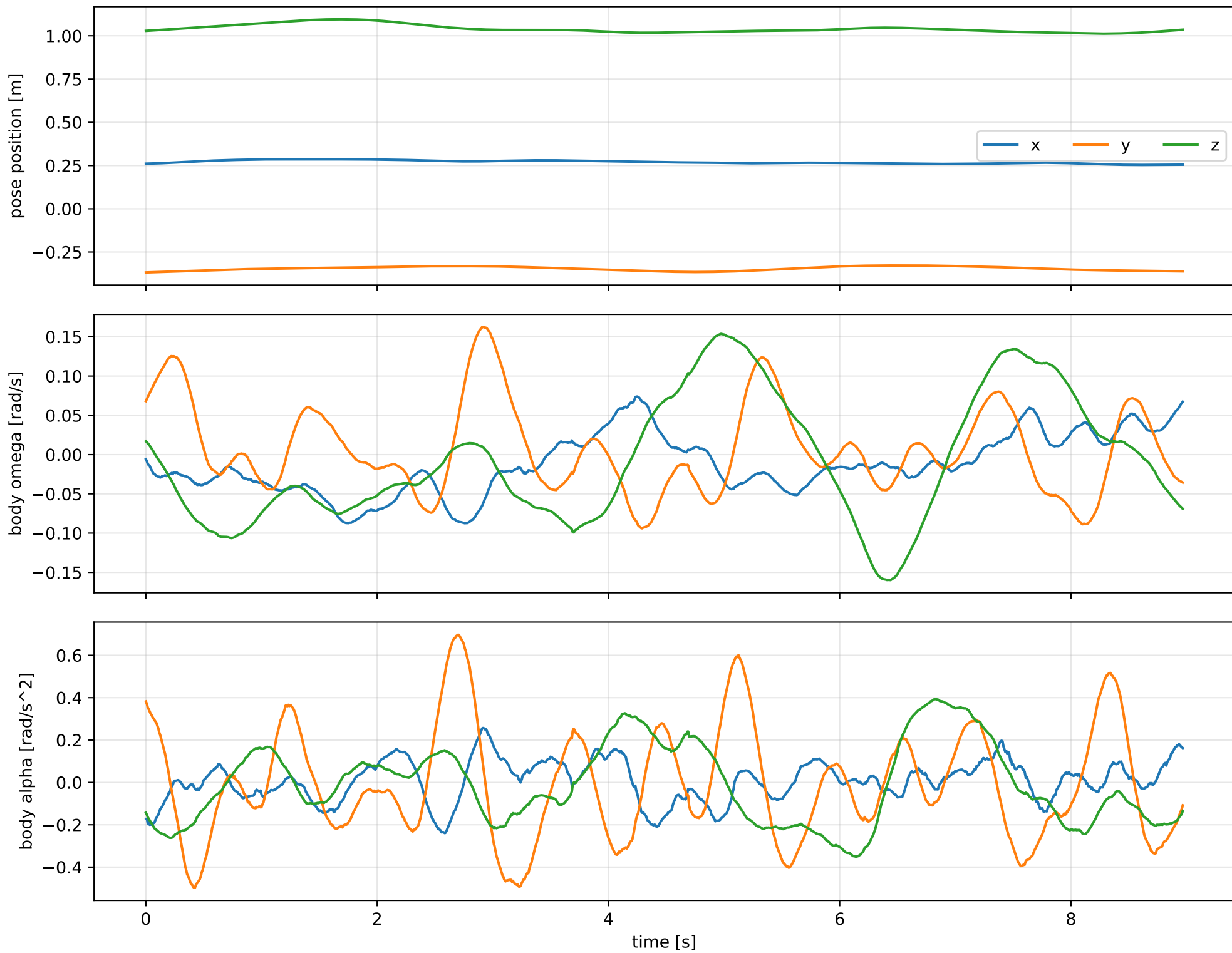
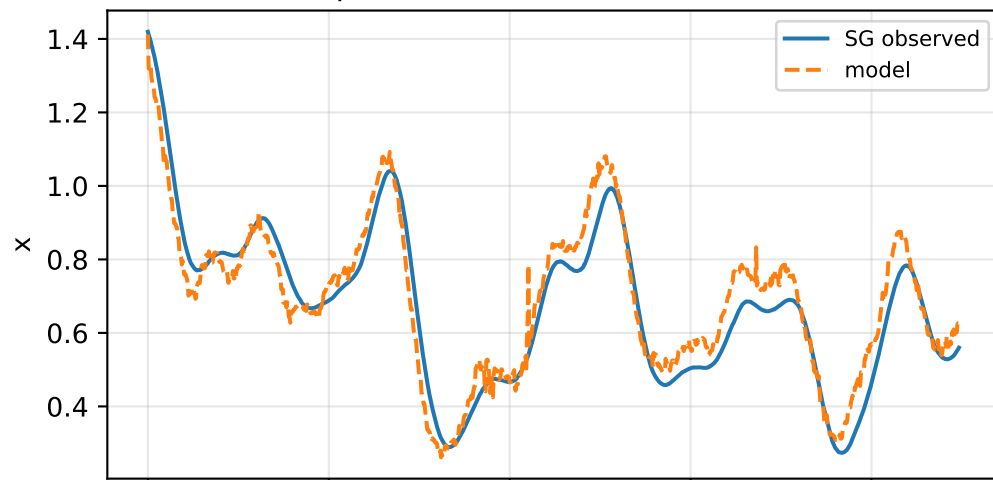


naive_all: SG trajectory and derived motion

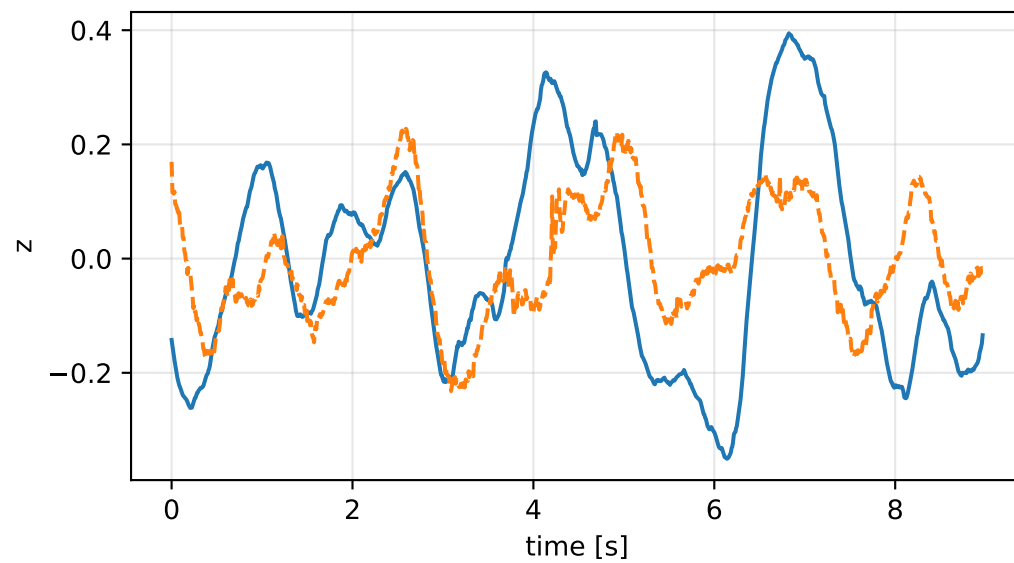
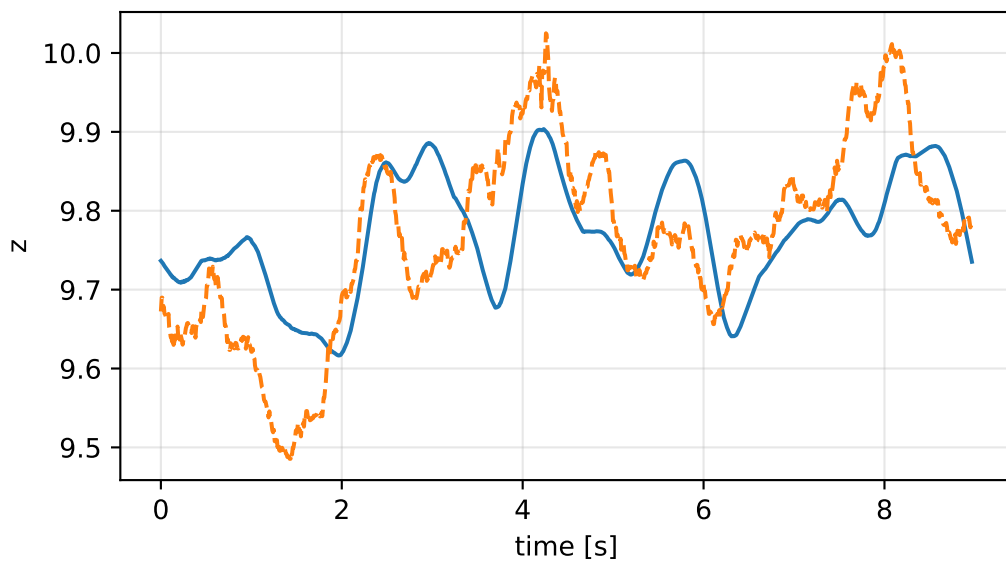
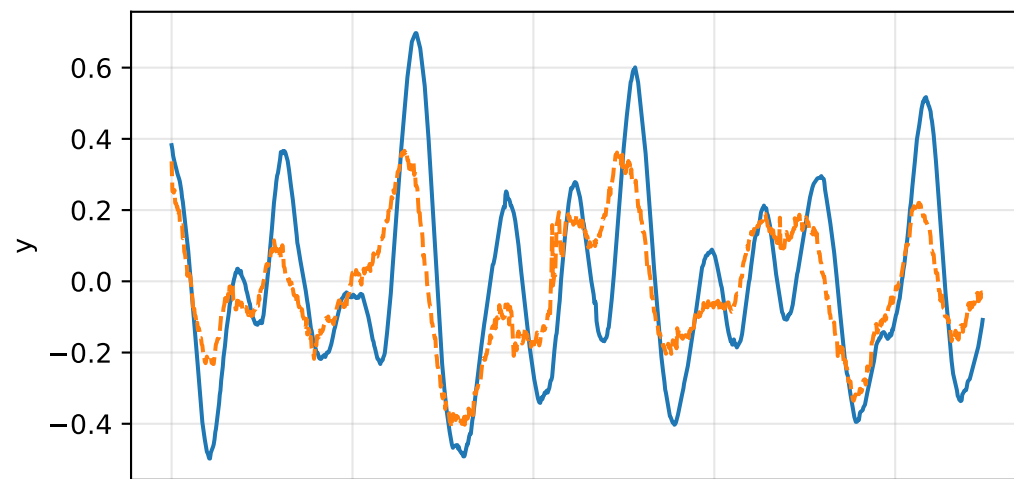
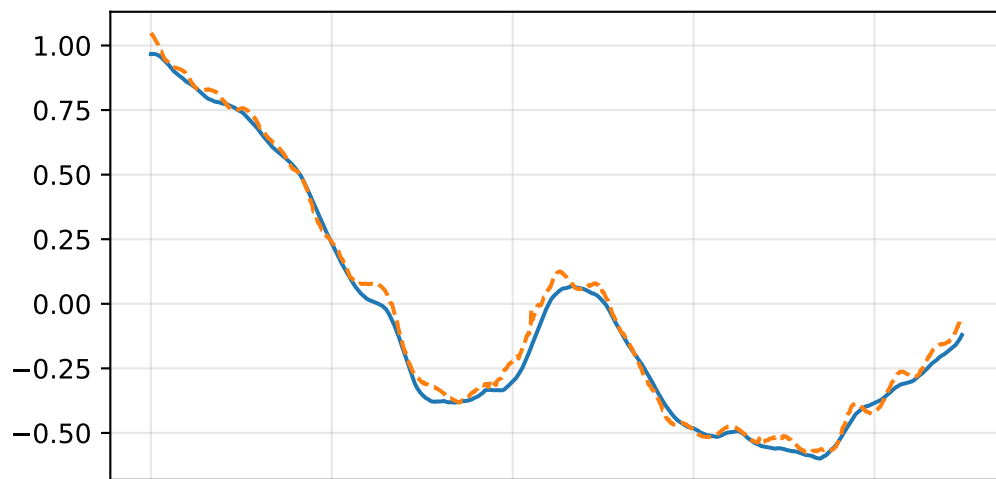
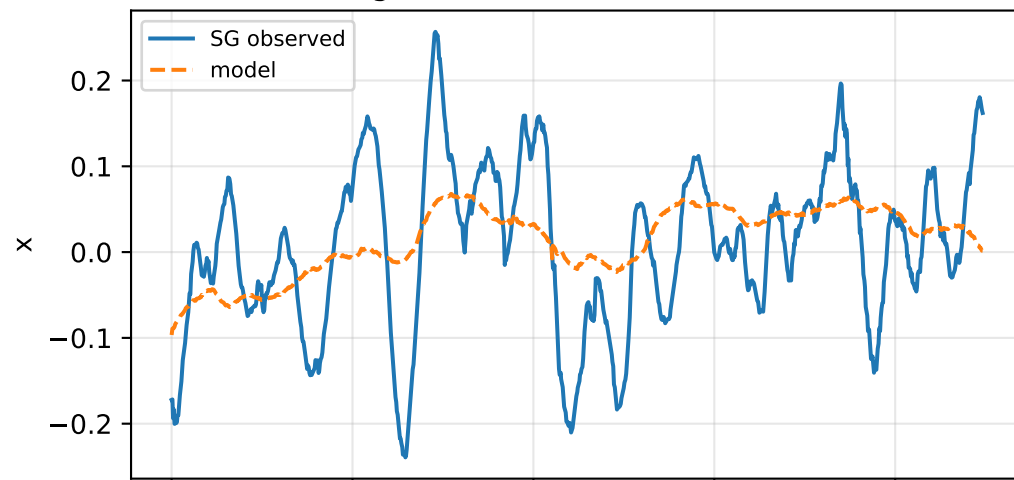


naive_all: acceleration residual objective

specific acceleration [m/s^2]

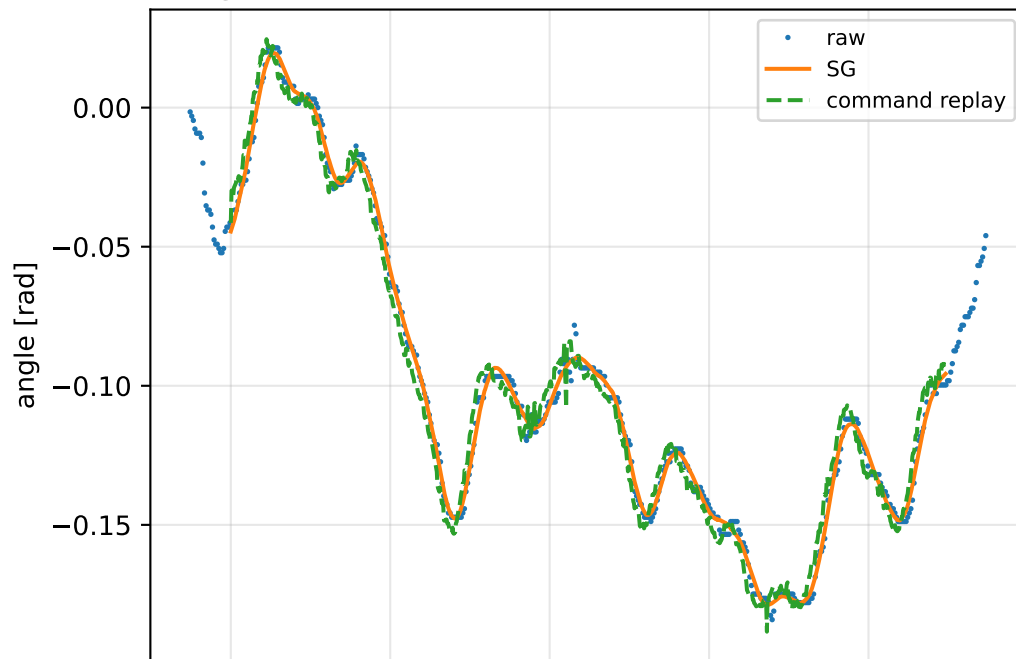


angular acceleration [rad/s^2]

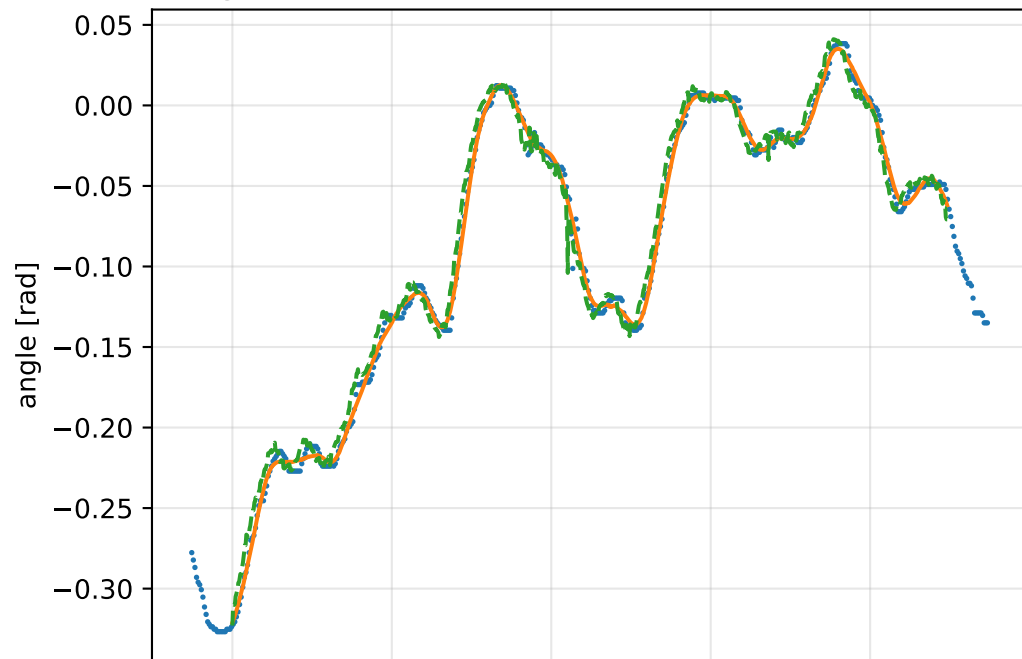


naive_all: gimbal measurement audit (objective=command_replay)

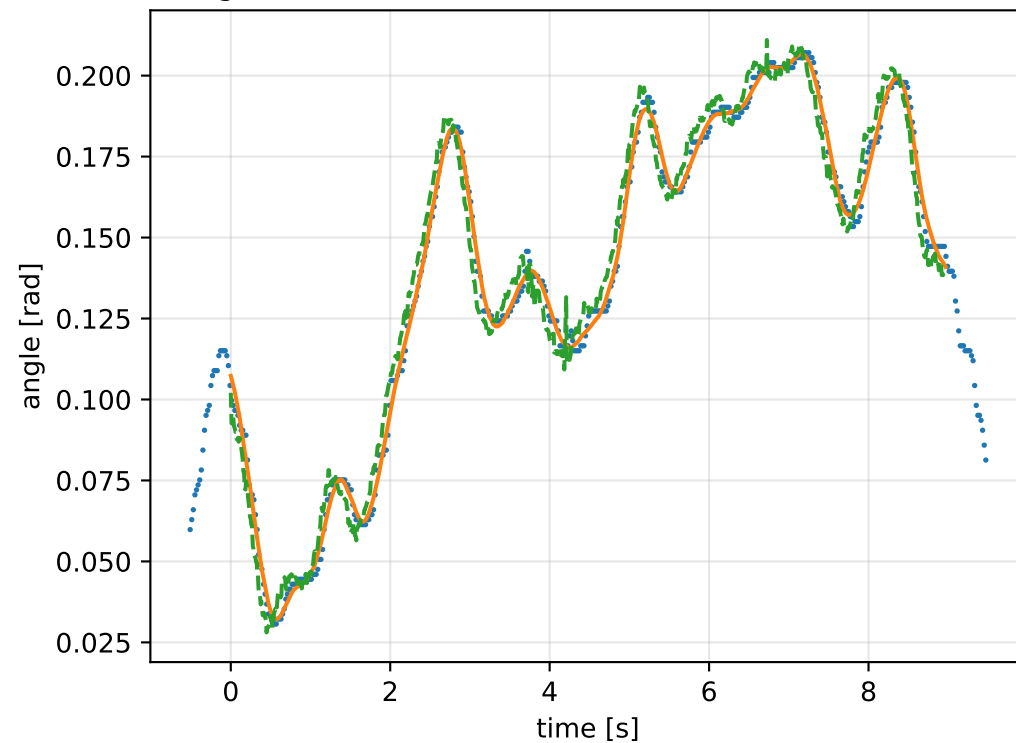
gimbal 1: RMSE=0.006345, max=0.01494 rad



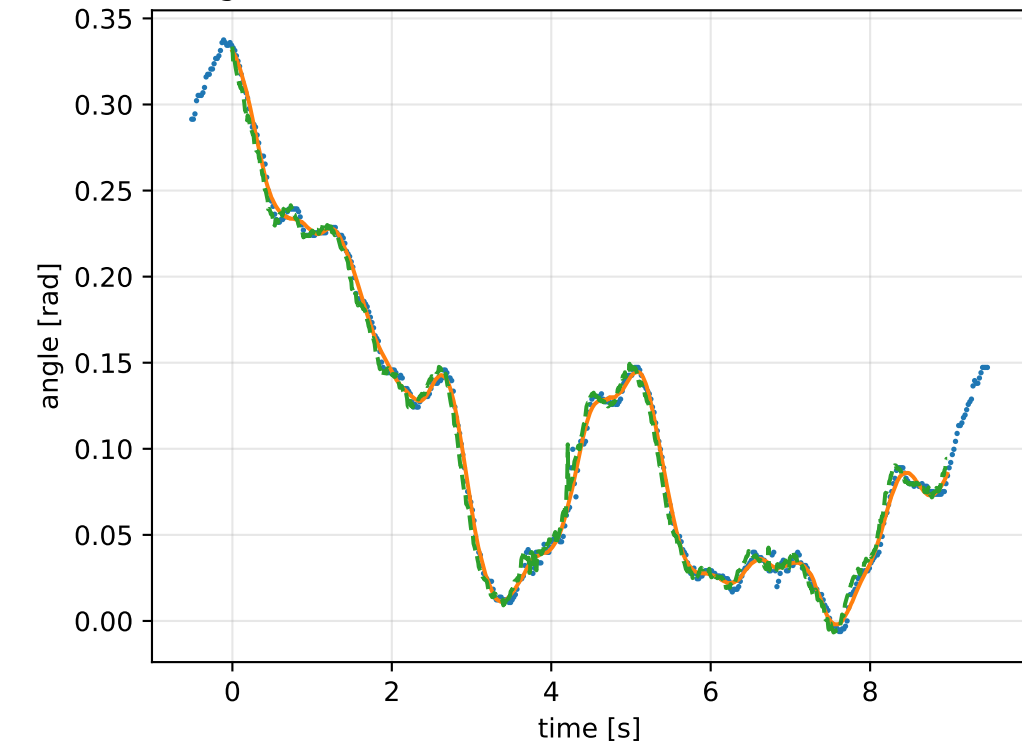
gimbal 2: RMSE=0.008835, max=0.04462 rad



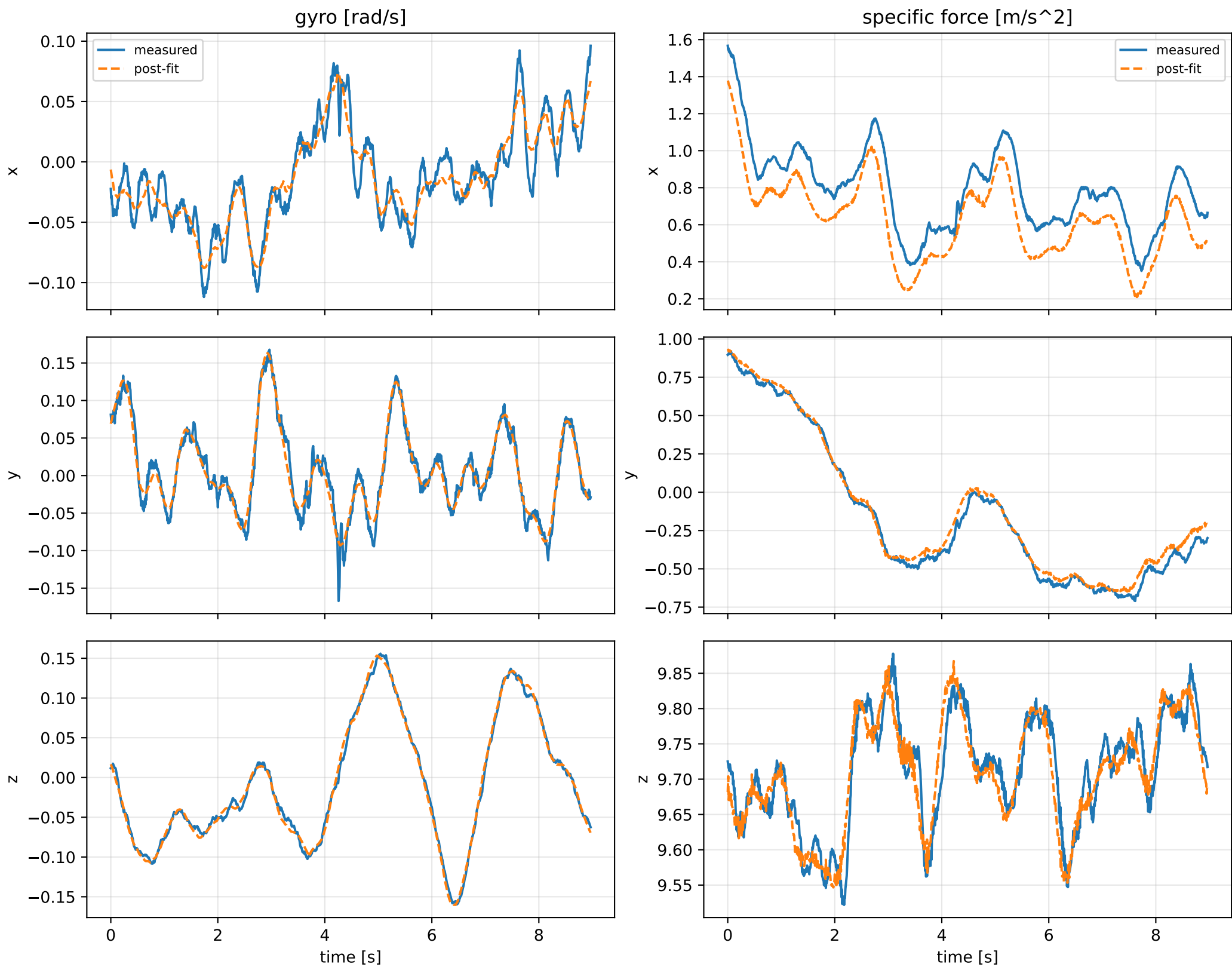
gimbal 3: RMSE=0.007019, max=0.01611 rad



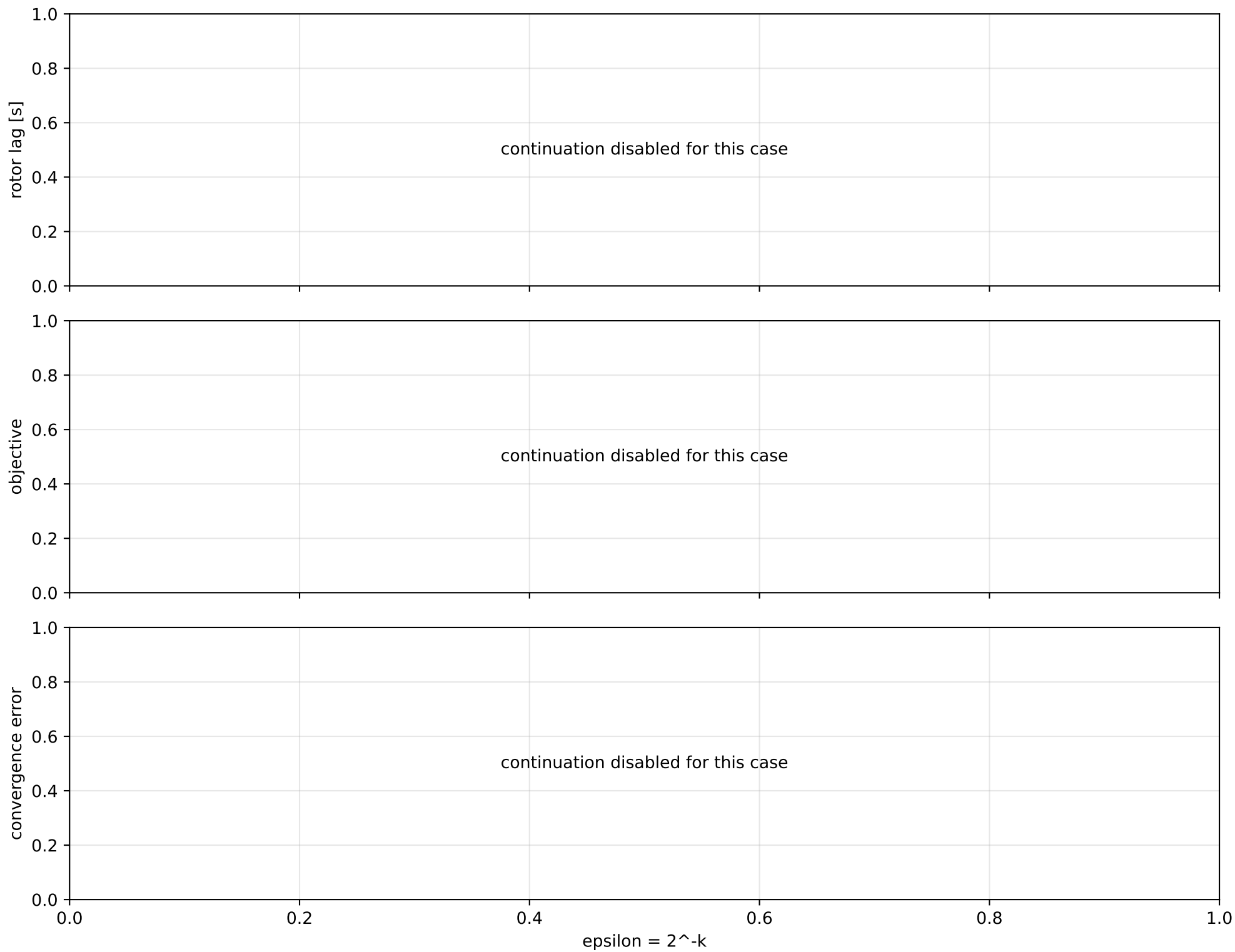
gimbal 4: RMSE=0.006321, max=0.03642 rad



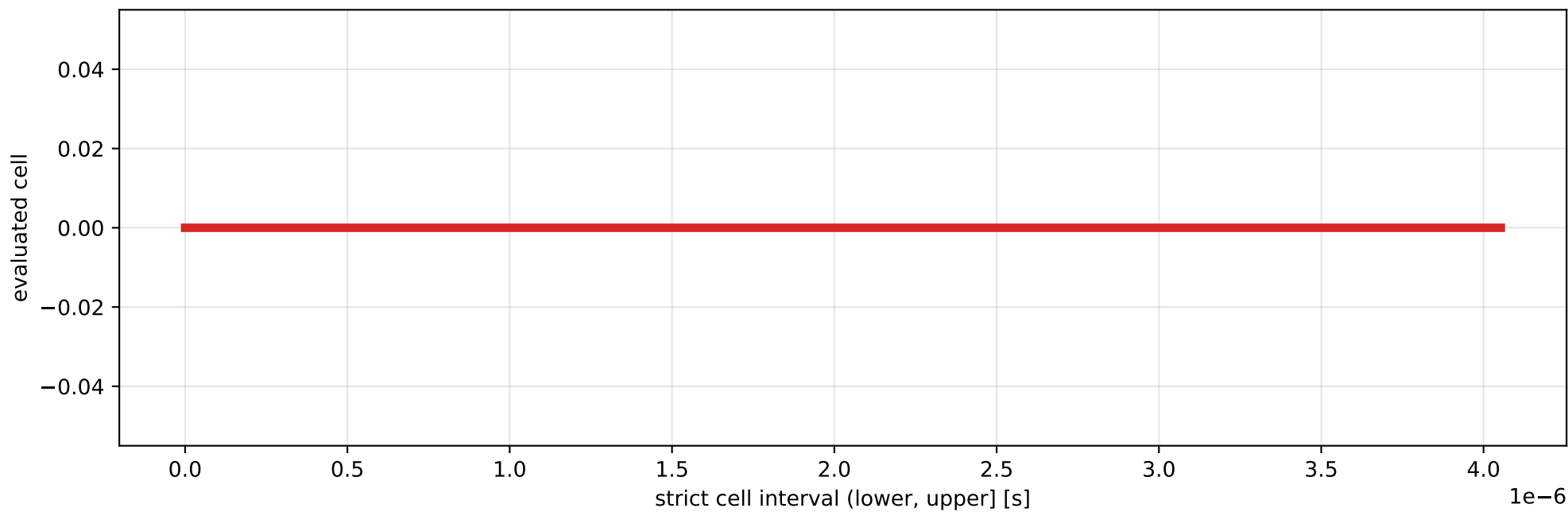
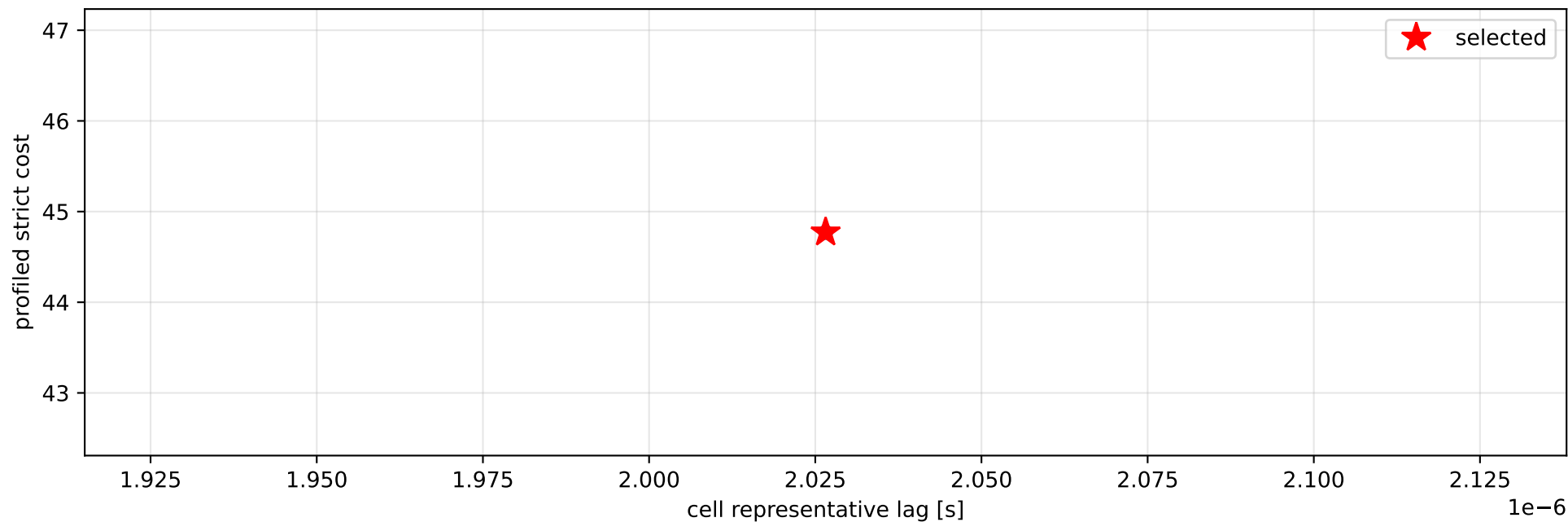
naive_all: IMU comparison (diagnostic only)



naive_all: smooth-to-strict continuation



naive_all: exact strict-ZOH lag cells



selected (0, 4.05311584e-06] s; final smooth 0 s; data boundary=False

naive_all: parameter estimate

Case: naive_all

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.71193009

inertia [kg m²]:

```
[[ 4.10682165  0.13793362  0.25282286]
 [ 0.13793362  1.38088849 -1.71650609]
 [ 0.25282286 -1.71650609  3.01657208]]
```

CoG body [m]: [-0.1229462 -0.05848602 -0.17657104]

force effectiveness: [1.00286604 0.48221458 0.1522316 0.69168136]

scale-free J/m [m²]:

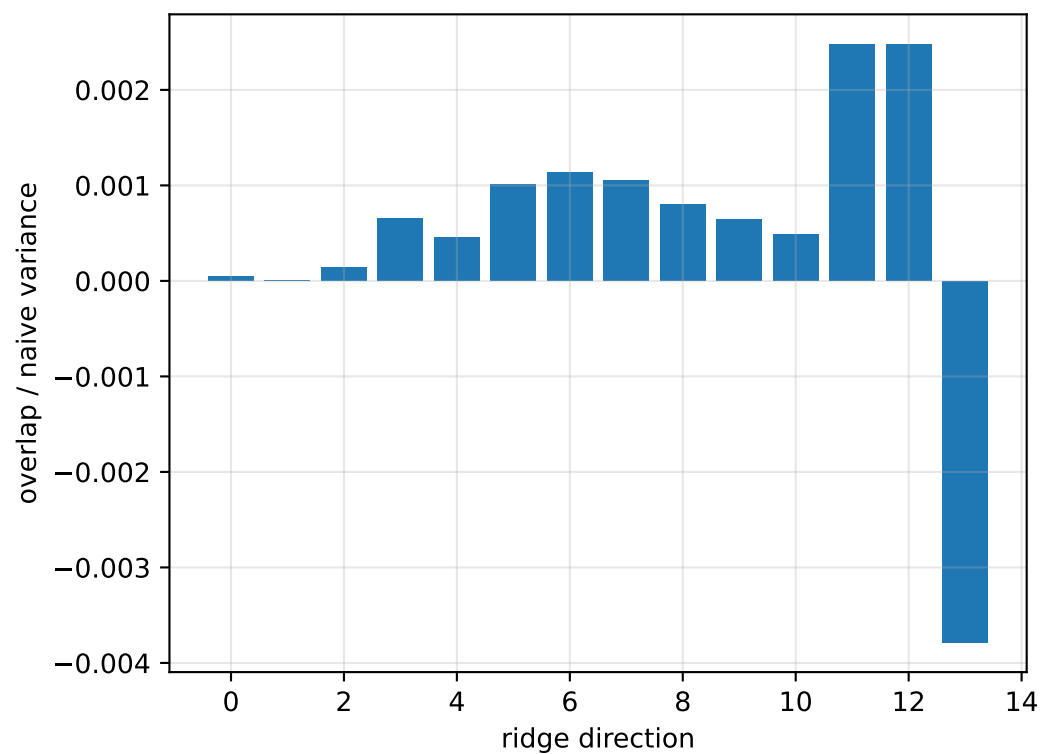
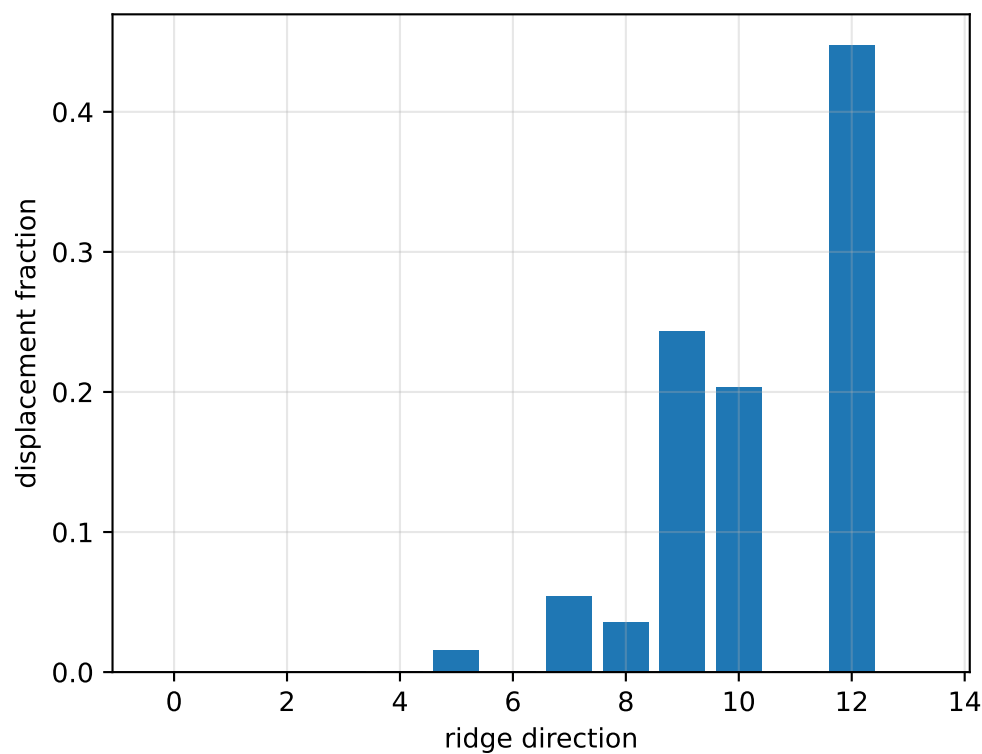
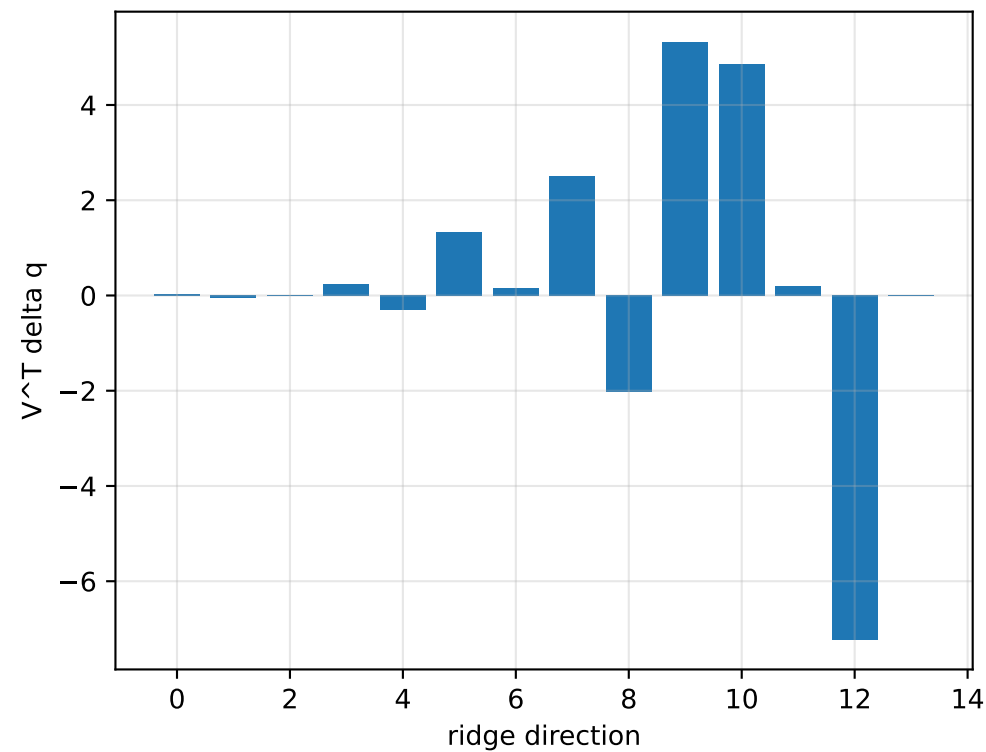
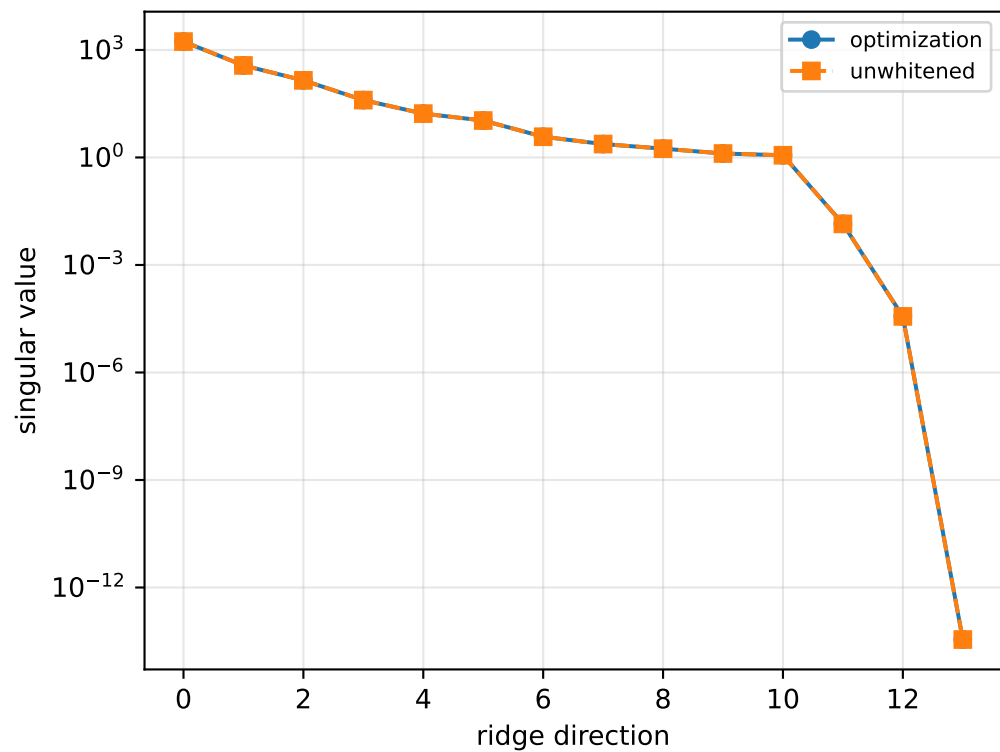
```
[[ 2.39894239  0.080572  0.14768294]
 [ 0.080572  0.80662668 -1.00267301]
 [ 0.14768294 -1.00267301  1.76208836]]
```

scale-free f/m: [0.58581016 0.2816789 0.08892396 0.40403599]

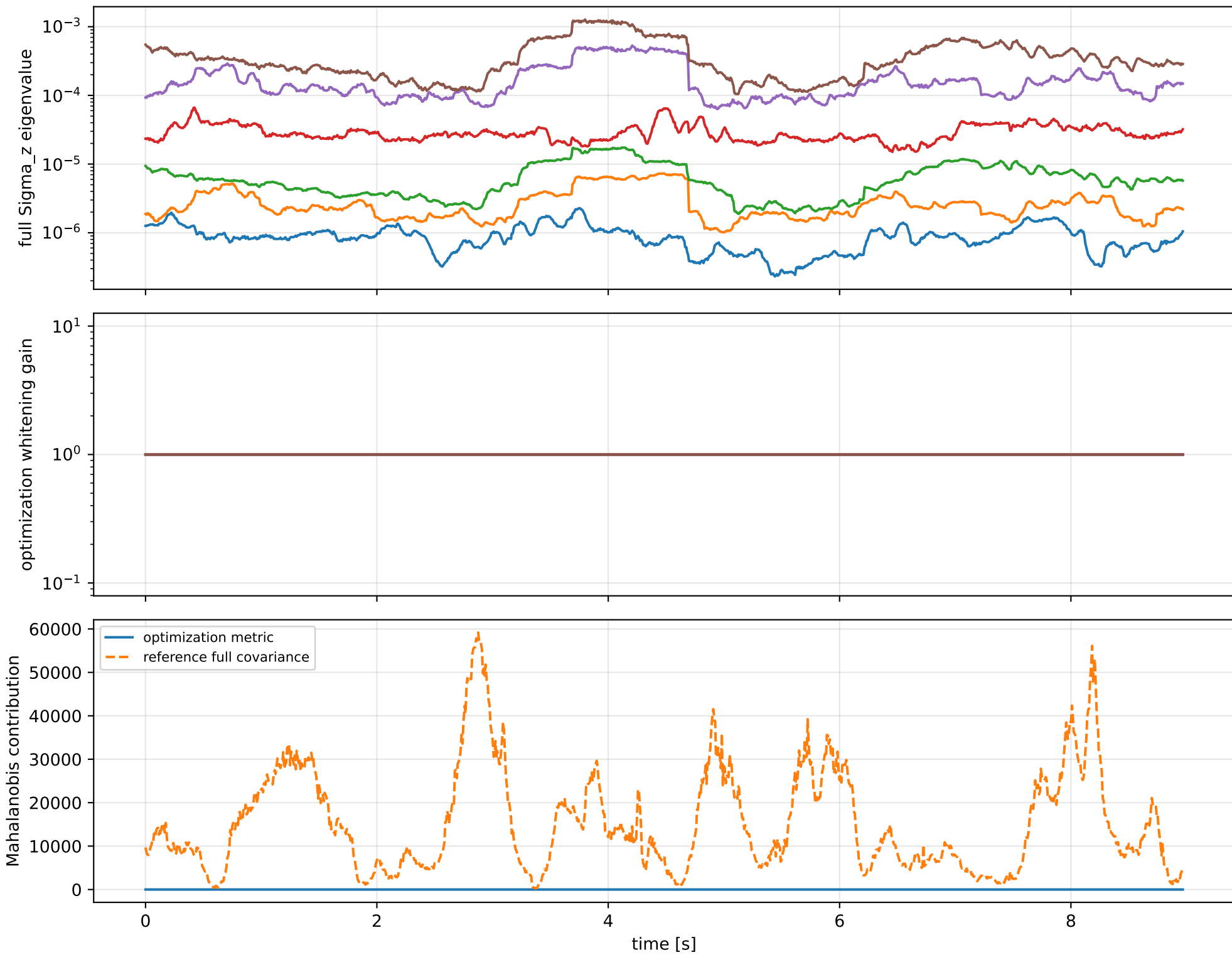
rotor lag cell: (0, 4.05311584e-06] s

representative: 0 s

naive_all: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=5.07\text{e-}14$)



naive_all: optimization vs reference covariance



naive_all: wrench / closure (force max 5.19e-15, torque max 4.44e-16)

